

# Homework 2

May 25, 2025

## 0.0.1 Problem 1.

```
[1]: from math import comb
import numpy as np
from scipy.stats import norm
np.random.seed(42)
```

```
[2]: T = [0.225, 0.262, 0.217, 0.240, 0.230, 0.229, 0.235, 0.217]
S = [0.209, 0.205, 0.196, 0.210, 0.202, 0.207, 0.224, 0.223, 0.220, 0.201]

mean_T = np.mean(T)
mean_S = np.mean(S)
var_T = np.var(T, ddof=1)
var_S = np.var(S, ddof=1)
n_T = len(T)
n_S = len(S)

se_diff = np.sqrt(var_T/n_T + var_S/n_S)
Z = (mean_T - mean_S) / se_diff
p_value = 2 * (1 - norm.cdf(abs(Z)))
Z_crit = norm.ppf(0.975)

ci_low = (mean_T - mean_S) - Z_crit * se_diff
ci_high = (mean_T - mean_S) + Z_crit * se_diff

print(f"p-value: {p_value:.4f} < 0.05 -> The average proportion of words in_
↳Twain's essays is different from that of Snodgrass.")
print(f"95% confidence interval: ({ci_low:.4f}, {ci_high:.4f})")
```

p-value: 0.0002 < 0.05 -> The average proportion of words in Twain's essays is different from that of Snodgrass.  
95% confidence interval: (0.0104, 0.0339)

```
[3]: combined = np.array(T + S)
perm_diffs = []

for _ in range(10_000):
    np.random.shuffle(combined)
    perm_T = combined[:n_T]
```

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perm_S = combined[n_T:]
perm_diffs.append(np.mean(perm_T) - np.mean(perm_S))

p_value = np.mean(np.abs(perm_diffs) >= abs(mean_T - mean_S))

print(f"P-value: {p_value:.4f} < 0.05")

```

P-value: 0.0008 < 0.05

## 0.0.2 Problem 2.

```

[4]: placebo = (80, 45)
p_values = []
drugs = {
    "Chlorpromazine": (75, 26),
    "Dimenhydrinate": (85, 52),
    "Pentobarbital (100 mg)": (67, 35),
    "Pentobarbital (150 mg)": (85, 37)
}

def fisher_exact_test(a, b, c=45, d=35):
    n = a + b + c + d
    row1 = a + b
    row2 = c + d
    col1 = a + c
    col2 = b + d

    def hypergeom_prob(x):
        return comb(col1, x) * comb(col2, row1 - x) / comb(n, row1)

    p_observed = hypergeom_prob(a)
    p_value = 0.0
    for i in range(max(0, row1 - col2), min(row1, col1) + 1):
        p = hypergeom_prob(i)
        if p <= p_observed:
            p_value += p

    odds_ratio = (a * d) / (b * c)
    return odds_ratio, p_value

print(f"{'Drug':37} {'Odds Ratio':12} {'P-value':12}")
for drug, (total, a) in drugs.items():
    b = total - a
    odds_ratio, p_value = fisher_exact_test(a, b)
    p_values.append(p_value)
    print(f"{drug:30} {odds_ratio:12.3f} {p_value:12.3f}")

```

| Drug                   | Odds Ratio | P-value |
|------------------------|------------|---------|
| Chlorpromazine         | 0.413      | 0.010   |
| Dimenhydrinate         | 1.226      | 0.531   |
| Pentobarbital (100 mg) | 0.851      | 0.740   |
| Pentobarbital (150 mg) | 0.600      | 0.120   |